

Deepak Mallampati

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Kent, OH (US-based, F-1 OPT, open to sponsorship)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) who ships GenAI proof points in days - LLM applications, RAG pipelines, agentic LLM workflows, and Stable Diffusion + ControlNet + LoRA diffusion-model work. Solo-shipped Manga Lens to the Chrome Web Store with four AI vision providers (Claude, GPT-4o mini, GPT-4.1 Nano, Ollama). FastAPI + React/TypeScript + Docker production discipline with ~30% post-deployment defect reduction; Healthcare RAG ~35% retrieval precision; agentic LLM workflows >30% hallucination reduction. Comfortable embedding with customer teams and translating technical limitations to non-technical stakeholders.

SKILLS

Forward Deployed Engineering • GenAI Application Delivery • LLMs & RAG & Agents • Stable Diffusion / ControlNet / LoRA • Multi-Provider GenAI APIs • FastAPI / React / Docker • Customer-Facing Delivery • CI/CD & Production Reliability

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React (Vite/Tailwind), Node.js (Chrome MV3), SQL (T-SQL, PostgreSQL, SQLite), C++ (Arduino), PHP

LLM & GenAI: LLM application development, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search, responsible AI

Diffusion Models / Multimodal: Stable Diffusion fine-tuning, LoRA, ControlNet, OpenPose/MediaPipe, YOLOv8, OpenCV, Hugging Face Diffusers, video analytics

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex

Integration & APIs: RESTful APIs, multi-provider API orchestration (Claude/OpenAI/Ollama, Perplexity Sonar + Replicate), JSON contracts, webhooks

Infra & DevOps: Docker, Jenkins CI/CD, Grafana, MLOps/LLMOps, structured logging, cloud-ready deployment

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - USA (Healthcare Technology, AI Consulting)

- Built **RAG systems** for clinical knowledge retrieval and documentation search; recursive semantic chunking + transformer embeddings; ~35% contextual retrieval precision gain, >30% irrelevant retrieval drop, >90% grounded-response alignment.
- Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning + tool discipline + grounding; agent response stability +~25%, hallucination reduction >30%.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, structured logging + load simulation; post-deployment defects ~30%.
- Wrote stakeholder-facing system-limitation docs alongside every release - the artifact non-technical clinicians and ops rely on.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India

- Built **CI/CD pipelines with Jenkins**; deployment errors ~30%, release cycle -35-40%.
- Designed performance dashboards (SQL Server DMVs + Grafana); incident recurrence ~25%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Hyderabad, India

- Built a **transformer-based video summarization** system; review time -60-70%; ~85% highlight alignment.
- Implemented **document Q&A with RAG**; hallucinated outputs ~30%, answer accuracy >85%.

PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation (shipped, Chrome Web Store)

Full-stack browser extension built independently - Manifest V3, content scripts, service workers. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local) with per-provider payload handling. Multi-section capture pipeline; 7-day cache; 29 site configs; privacy policy; narrowed host permissions. **Direct multi-provider GenAI integration proof.**

Agentic Pixel Character Synthesis & Animation Engine

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis. **Stable Diffusion fine-tuning + LoRA** for identity consistency, **ControlNet + OpenPose/MediaPipe** for pose conditioning, latent-space consistency for animation smoothness, and an **LLM-based agent orchestrator**. PyTorch + HF Diffusers + FastAPI + Docker.

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating multimodal APIs (Perplexity Sonar + Replicate).

Intermediate structured prompt transformation layers improved contextual alignment **~30%** and first-pass image success rate **~25-30%**.

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline with **schema-validated JSON contracts between agents**. RAG-grounded CPT/ICD validation; ANN duplicate detection; explainable risk scoring with audit-ready reasoning traces.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025